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STATISTICAL SIGNAL PROCESSING - *Prova scritta del 16/02/2023*

We observe N samples of a realization of the random process $X[n] = S[n] + W[n]$, for n=0, 1,., N-1, with N>>1. $S[n] = A\cos(2\pi F_0 n)$ is the signal of interest (SOI), A is the amplitude of the SOI that is modelled as a random parameter; in particular, it is modelled as a Gaussian random variable with known mean $\eta_A = 0$ and variance σ_A^2, the signal frequency F_0 is a priori known, $W[n]$ is additive white Gaussian noise (AWGN), independent of A, with known Power Spectral Density (PSD) $S_W\left(e^{j2\pi f}\right) = \sigma_W^2$.

(1) Derive the mean function $\eta_X[n] = E\{X[n]\}$ and the Autocorrelation Function (ACF) $R_X[n,m] = E\{X[n]X[n+m]\}$ of the process $X[n]$ and determine whether it is w.s.s. or not.

(2) Derive the signal to noise power ratio (SNR) $\gamma \triangleq E\left\{\|\mathbf{S}\|^2\right\} \Big/ E\left\{\|\mathbf{W}\|^2\right\}$, where **S** and **W** are the vectors containing the N samples of signal and noise, respectively. Then, derive the joint probability density function (pdf) $f_{X,A}(\mathbf{x}, a)$ of the data vector $\mathbf{X} \triangleq [X[0] \quad X[1] \quad \cdots \quad X[N-1]]^T$ and of the random parameter A (this pdf can be expressed either in scalar form or in vector form).

(3) Derive the Maximum A-Posteriori (MAP) estimator of the amplitude A and express it in terms of the ML estimate of A and of the SNR γ.

(4) Derive the bias and the Mean Square Error (MSE) of the MAP estimator and express the MSE in terms of the MSE of the ML estimate of A and of the SNR γ. Plot the MSE as a function of the sample size N and establish whether the MAP estimator is unbiased and consistent or not.

(5) Derive the Bayesian Cramér-Rao lower bound (BCRB) on the estimate of A and establish whether the MAP estimator is efficient or not.

Solution

(1) Mean function and ACF:

$$\eta_X[n] = E\{X[n]\} = E\{A\cos(2\pi F_0 n) + W[n]\} = E\{A\}\cos(2\pi F_0 n) = 0$$

$$\begin{aligned} R_X[n,m] &= E\{X[n]X[n+m]\} = E\{(A\cos(2\pi F_0 n) + W[n])(A\cos(2\pi F_0(n+m)) + W[n+m])\} \\ &= E\{A^2\}\cos(2\pi F_0 n)\cos(2\pi F_0(n+m)) + E\{W[n]W[n+m]\} \\ &= \sigma_A^2 \cos(2\pi F_0 n)\cos(2\pi F_0(n+m)) + \sigma_W^2 \delta[m] \end{aligned}$$

where we used the fact that A and $W[n]$ are independent and zero mean. Since the ACF is not a function of only the lag m, but it also depends on the time n, the process is not wide sense stationary.

(2) SNR and joint PDF:

$$E\{\|\mathbf{S}\|^2\} = E\{\mathbf{S}^T\mathbf{S}\} = E\{A^2\mathbf{c}^T\mathbf{c}\} = E\{A^2\}\sum_{n=0}^{N-1}\cos^2(2\pi F_0) \cong \sigma_A^2\frac{N}{2},\ E\{\|\mathbf{W}\|^2\} = E\{\mathbf{W}^T\mathbf{W}\} = N\sigma_W^2,$$

where $\mathbf{S} = A\mathbf{c}$, $\mathbf{c} \triangleq [1 \quad \cos(2\pi F_0) \quad \cdots \quad \cos(2\pi F_0(N-1))]^T$

and $\|\mathbf{c}\|^2 = \mathbf{c}^T\mathbf{c} = \sum_{n=0}^{N-1}\cos^2(2\pi F_0) \cong \frac{N}{2}$ if $N \gg 1$.

$$\gamma = E\{\|\mathbf{S}\|^2\} \big/ E\{\|\mathbf{W}\|^2\} = \frac{\sigma_A^2}{2\sigma_W^2}.$$

$f_{X,A}(\mathbf{x},a) = f_{X|A}(\mathbf{x}|a)f_A(a)$, where $A \in N(0,\sigma_A^2)$

$X_n|A \in N(A\cos(2\pi F_0 n), \sigma_W^2)$, $\{X_n|A\}_{n=0}^{N-1}$ conditionally independent $\Rightarrow$ $\mathbf{X}|A \in N(A\mathbf{c}, \sigma_W^2\mathbf{I})$

The pdf in scalar format is:

$$\begin{aligned} f_{X,A}(\mathbf{x},a) &= f_{X|A}(\mathbf{x}|a)f_A(a) = \prod_{n=0}^{N-1} f_{X|A}(x_n|a)f_A(a) \\ &= \left(\frac{1}{\sqrt{2\pi\sigma_W^2}}\right)^N \exp\left(-\frac{1}{2\sigma_W^2}\sum_{n=0}^{N-1}(x_n - a\cos(2\pi F_0 n))^2\right)\cdot\frac{1}{\sqrt{2\pi\sigma_A^2}}\exp\left(-\frac{a^2}{2\sigma_A^2}\right) \end{aligned}$$

In vector format:

$$f_{X,A}(\mathbf{x},a) = f_{X|A}(\mathbf{x}|a)f_A(a) = \frac{1}{(2\pi\sigma_W^2)^{N/2}}\exp\left(-\frac{1}{2\sigma_W^2}(\mathbf{x}-a\mathbf{c})^T(\mathbf{x}-a\mathbf{c})\right)\cdot\frac{1}{\sqrt{2\pi\sigma_A^2}}\exp\left(-\frac{a^2}{2\sigma_A^2}\right)$$

(3) MAP estimator of A:

$$\hat{A}_{MAP} = \arg\max_a \left(\ln f_{X,A}(\mathbf{x},a)\right) = \arg\max_a \left(\ln f_{X|A}(\mathbf{x}|a) + \ln f_A(a)\right)$$

$$\begin{aligned}
\ln f_{X|A}(\mathbf{x}|a) + \ln f_A(a) &= \kappa - \frac{1}{2\sigma_W^2}(\mathbf{x}-a\mathbf{c})^T(\mathbf{x}-a\mathbf{c}) - \frac{a^2}{2\sigma_A^2} \\
&= \kappa - \frac{1}{2\sigma_W^2}\mathbf{x}^T\mathbf{x} - \frac{a^2}{2\sigma_W^2}\mathbf{c}^T\mathbf{c} + \frac{1}{2\sigma_W^2}a\mathbf{c}^T\mathbf{x} + \frac{1}{2\sigma_W^2}a\mathbf{x}^T\mathbf{c} - \frac{a^2}{2\sigma_A^2} \\
&= \kappa - \frac{1}{2\sigma_W^2}\mathbf{x}^T\mathbf{x} - \frac{Na^2}{4\sigma_W^2} + \frac{a}{\sigma_W^2}\left(\mathbf{c}^T\mathbf{x}\right) - \frac{a^2}{2\sigma_A^2}
\end{aligned}$$

$$\frac{d}{da}\left(\ln f_{X|A}(\mathbf{x}|a) + \ln f_A(a)\right) = -\frac{Na}{2\sigma_W^2} + \frac{N}{2\sigma_W^2}\left(\frac{2}{N}\mathbf{c}^T\mathbf{x}\right) - \frac{a}{\sigma_A^2} = \frac{N}{2\sigma_W^2}\hat{A}_{ML} - \left(\frac{N}{2\sigma_W^2} + \frac{1}{\sigma_A^2}\right)a = 0$$

$$\hat{A}_{MAP} = \left(\frac{N}{2\sigma_W^2} + \frac{1}{\sigma_A^2}\right)^{-1}\left(\frac{N}{2\sigma_W^2}\hat{A}_{ML}\right) = \frac{2\sigma_A^2\sigma_W^2}{\sigma_A^2 N + 2\sigma_W^2}\left(\frac{N}{2\sigma_W^2}\hat{A}_{ML}\right)\frac{\sigma_A^2 N}{\sigma_A^2 N + 2\sigma_W^2}\hat{A}_{ML} = \frac{N\gamma}{1+N\gamma}\hat{A}_{ML}$$

$$\hat{A}_{MAP} = \alpha\hat{A}_{ML}, \quad \text{where } \hat{A}_{ML} = \frac{\mathbf{c}^T\mathbf{x}}{\mathbf{c}^T\mathbf{c}} = \frac{2}{N}\mathbf{c}^T\mathbf{x} = \frac{2}{N}\sum_{n=0}^{N-1} X[n]\cos(2\pi F_0), \quad \gamma = \frac{\sigma_A^2}{2\sigma_W^2} \text{ and } \alpha \triangleq \frac{N\gamma}{1+N\gamma}$$

Note that the ML estimator of A is the well-known matched filter (matched to $\mathbf{c}$). Hence, the estimator is **linear**. Since the estimate is a linear function of the Gaussian distributed data vector $\mathbf{X}$, the estimate is Gaussian distributed, and consequently the estimation error is Gaussian distributed, with mean and variance equal to the bias and the MSE, respectively.

(4) Bias and MSE of the MAP estimator of A:

$$E\left\{\hat{A}_{MAP}\right\} = E\left\{\alpha \cdot \frac{2}{N}\sum_{n=0}^{N-1} X[n]\cos(2\pi F_0)\right\} = \frac{2\alpha}{N}\sum_{n=0}^{N-1} E\left\{X[n]\right\}\cos(2\pi F_0) = 0$$

$\Rightarrow \quad E\left\{A - \hat{A}_{MAP}\right\} = E\left\{A\right\} - E\left\{\hat{A}_{MAP}\right\} = 0$. Hence, the estimator is **unbiased**.

$$
\begin{aligned}
MSE\left\{\hat{A}_{MAP}\right\} &= E\left\{\left(A-\hat{A}_{MAP}\right)^2\right\} = E\left\{\left(A-\frac{2\alpha}{N}\mathbf{c}^T\mathbf{x}\right)^2\right\} = E\left\{\left(A-\frac{2\alpha}{N}\mathbf{c}^T\left(A\mathbf{c}+\mathbf{W}\right)\right)^2\right\} \\
&= E\left\{\left(A-\frac{2\alpha A}{N}\mathbf{c}^T\mathbf{c}+\frac{2\alpha}{N}\mathbf{c}^T\mathbf{W}\right)^2\right\} = E\left\{\left((1-\alpha)A+\frac{2\alpha}{N}\mathbf{c}^T\mathbf{W}\right)^2\right\} \\
&= E\left\{\left((1-\alpha)A\right)^2\right\} + E\left\{\left(\frac{2\alpha}{N}\mathbf{c}^T\mathbf{W}\right)^2\right\} = (1-\alpha)^2 E\left\{A^2\right\} + \frac{4\alpha^2}{N^2}E\left\{\left(\mathbf{c}^T\mathbf{W}\right)^2\right\} \\
&= (1-\alpha)^2\sigma_A^2 + \frac{4\alpha^2}{N^2}\mathbf{c}^T E\left\{\mathbf{W}\mathbf{W}^T\right\}\mathbf{c} = (1-\alpha)^2\sigma_A^2 + \frac{4\alpha^2}{N^2}\cdot\frac{N}{2}\sigma_W^2 = (1-\alpha)^2\sigma_A^2 + \frac{2\alpha^2}{N}\sigma_W^2 \\
&= \left((1-\alpha)^2 N\gamma + \alpha^2\right)\frac{2\sigma_W^2}{N} = \alpha\frac{2\sigma_W^2}{N} = \alpha MSE\left\{\hat{A}_{ML}\right\}
\end{aligned}
$$

where we used the fact that A and $\mathbf{W}$ are independent and zero mean and the fact that $MSE\left\{\hat{A}_{ML}\right\} = \frac{2\sigma_W^2}{N}$. Hence, the PDF of the estimation error: $\varepsilon = A - \hat{A}_{MAP} \in N\left(0, \alpha\frac{2\sigma_W^2}{N}\right)$.

$\lim_{N\to\infty} MSE\left\{\hat{A}_{MAP}\right\} = \lim_{N\to\infty}\left(\alpha\frac{2\sigma_W^2}{N}\right) = 0,$ since $\lim_{N\to\infty}\alpha = 1$. Hence, the estimator is **consistent**.

(5) BCRB on the estimate of *A*:

$$
\frac{d^2 \ln f_{X,A}(\mathbf{x},a)}{da^2} = \frac{d^2}{da^2}\left(\ln f_{X|A}(\mathbf{x}|a) + \ln f_A(a)\right) = \frac{d}{da}\left[\frac{N}{2\sigma_W^2}\hat{A}_{ML} - \left(\frac{N}{2\sigma_W^2}+\frac{1}{\sigma_A^2}\right)a\right] = -\left(\frac{N}{2\sigma_W^2}+\frac{1}{\sigma_A^2}\right)
$$

$$
\begin{aligned}
BCRB(A) &= \frac{1}{-E\left\{\frac{d^2\ln f_{X,A}(\mathbf{x},a)}{da^2}\right\}} = \left(\frac{N}{2\sigma_W^2}+\frac{1}{\sigma_A^2}\right)^{-1} = \frac{2\sigma_A^2\sigma_W^2}{\sigma_A^2 N + 2\sigma_W^2} \\
&= \frac{\sigma_A^2 N}{\sigma_A^2 N + 2\sigma_W^2}\cdot\frac{2\sigma_W^2}{N} = \frac{\gamma N}{\gamma N + 1}\cdot\frac{2\sigma_W^2}{N} = \alpha\frac{2\sigma_W^2}{N} = MSE\left\{\hat{A}_{MAP}\right\}
\end{aligned}
$$

Hence, the estimator is **efficent**.